

A Certified Lower Bound for the Maximum Volume Wrappable by a Disk

Sho Hoshino

August 11, 2026

Abstract

For the closed disk $D_R \subset \mathbb{R}^2$, let

$$W(R) := \sup\{\text{Vol}(B) : B \text{ is wrappable by } D_R\}$$

in the 1-Lipschitz wrapping formalization of Nandakumar. We prove the explicit certified lower bound

$$W(R) \geq CR^3, \quad C = 0.17984349416291615\dots,$$

where C is an exact rational number. More precisely, we give a 17-vertex, 30-face embedded nonconvex polyhedral sphere and a continuous 1-Lipschitz map from D_R onto it. Two independent exact-arithmetic programs certify strict nonexpansion on all 30 source triangles, the topology of the quotient complex, all 435 triangle-pair intersections, nonconvexity, and the exact enclosed volume.

Nandakumar’s Question 1 asks only whether D_π can wrap some nonconvex solid of volume greater than $4\pi/3$. Under the stated 1-Lipschitz formalization, that question has a short soft affirmative answer: an oblate spheroid with equatorial radius $23/20$ and polar radius $4/5$ has volume greater than the unit ball and north–south meridian length strictly less than π ; a sufficiently small inward rotational notch preserves both strict inequalities while making the body nonconvex. A simple radial lemma then gives a 1-Lipschitz wrap by D_π . Thus the point of the certified polyhedral construction is not existence of a nonconvex improvement over the sphere, but the substantially stronger quantitative lower bound above. For comparison, the classical Mylar balloon is the sharp convex body-of-revolution benchmark with coefficient $0.152315527\dots$, so our certified coefficient is about 18.07% larger.

1 Introduction

Nandakumar recently formulated a volume-maximization problem for wrapping three-dimensional solids by a planar sheet that may be folded or wrinkled but not stretched or torn [1]. In his formalization, a planar region D wraps a compact connected solid when its boundary surface is realized by a 1-Lipschitz image of D , with the boundary of D glued to itself in the image. For the disk D_R of radius R , it is therefore natural to write

$$W(R) := \sup\{\text{Vol}(B) : B \text{ is wrappable by } D_R\}. \tag{1}$$

Scaling gives $W(R) = R^3W(1)$ whenever the supremum is interpreted in this fixed formalization, so the main quantitative problem is to understand the dimensionless constant $W(1)$.

Nandakumar’s main conjecture is explicitly nonconvex: every wrappable convex solid should be beaten in volume by some wrappable nonconvex solid [1, Claim 1]. His Question 1 fixes the unit

ball as a test case and asks whether D_π can wrap a nonconvex solid of volume greater than $4\pi/3$ [1, Question 1]. A subsequent MathOverflow remark by Nandakumar already records convex wrappable bodies with volume larger than the unit sphere [2]. Under the stated 1-Lipschitz formalization, one can go one step further: as shown in section 2, a tiny inward dent of a suitably chosen oblate spheroid gives a short elementary affirmative answer to Question 1. Thus Question 1 itself is not the quantitative difficulty addressed here.

The purpose of this note is to give a substantially stronger explicit lower bound for (1). The construction is finite and completely specified: we give the source triangulation, target coordinates, and piecewise-affine map. The proof then reduces to a finite exact certificate. Two supplied implementations use different arithmetic and different triangle-intersection algorithms.

Our main result is the following.

Theorem 1 (Certified disk-wrapping lower bound). *Let $D_R \subset \mathbb{R}^2$ be the closed disk of radius $R > 0$. There exists a continuous 1-Lipschitz map*

$$F_R : D_R \longrightarrow \mathbb{R}^3$$

whose image is an embedded nonconvex polyhedral 2-sphere ∂B_R . The bounded solid B_R has volume

$$\text{Vol}(B_R) = CR^3, \quad (2)$$

where

[illegible]

In particular,

$$W(R) \geq CR^3 > 0.179843R^3.$$

The classical Mylar balloon provides a useful convex benchmark. Among convex bodies of revolution with intrinsic diameter R , it has the sharp volume

$$V_{\text{Mylar}}(R) = 0.152315527 \dots R^3,$$

and it is itself wrappable by the single disk D_R . Hence Theorem 1 improves this axisymmetric convex benchmark by about 18.07%. We do not claim that Mylar is optimal among all convex bodies wrappable by D_R , nor do we claim to determine $W(R)$.

The geometry of the certified witness was discovered by numerical search, but the theorem depends only on the final rational-coordinate certificate. Its boundary identification has one short pleat branch and otherwise consists of doubled seam segments; no physical analogy is needed in the proof.

2 A soft affirmative answer to Nandakumar's Question 1

We first isolate a simple observation that applies to convex and nonconvex surfaces of revolution alike.

Lemma 2 (Radial wrapping of a surface of revolution). *Let*

$$\gamma(s) = (\rho(s), z(s)), \quad 0 \leq s \leq L,$$

be a simple rectifiable arc in the half-plane $\varrho \geq 0$, parametrized by arclength, with

$$\varrho(0) = \varrho(L) = 0, \quad \varrho(s) > 0 \quad (0 < s < L).$$

Let S be the embedded surface obtained by revolving γ about the z -axis. If $L \leq R$, then D_R wraps the bounded solid enclosed by S in Nandakumar's 1-Lipschitz sense.

Proof. Set $q = L/R \leq 1$. In polar coordinates (r, θ) on D_R , define

$$F(r, \theta) = (\varrho(qr) \cos \theta, \varrho(qr) \sin \theta, z(qr)).$$

Since γ is parametrized by arclength,

$$\varrho'(s)^2 + z'(s)^2 = 1$$

for almost every s , so $|\varrho'| \leq 1$. Because $\varrho(0) = 0$,

$$0 \leq \varrho(s) \leq s.$$

Away from the rotation axis the pullback metric is

$$F^*g = q^2 dr^2 + \varrho(qr)^2 d\theta^2 \leq q^2(dr^2 + r^2 d\theta^2) \leq dr^2 + r^2 d\theta^2.$$

Integrating lengths of curves and using continuity at the axis gives $\text{Lip}(F) \leq q \leq 1$. The map is onto S , while the whole boundary circle $r = R$ is sent to the endpoint $\gamma(L)$ on the axis. Hence the disk boundary is self-glued and the image is the closed surface S . $\square$

Proposition 3 (A soft answer to Question 1). *A disk of radius π wraps a nonconvex solid of volume strictly greater than $4\pi/3$.*

Proof. Consider the oblate spheroid with equatorial radius

$$a = \frac{23}{20}$$

and polar radius

$$c = \frac{4}{5}.$$

Its volume is

$$V_0 = \frac{4\pi}{3} a^2 c = \frac{4\pi}{3} \frac{529}{500} > \frac{4\pi}{3}.$$

A north-south generating meridian is

$$\gamma_0(t) = (a \sin t, c \cos t), \quad 0 \leq t \leq \pi.$$

If L_0 denotes its length, Cauchy-Schwarz gives

$$\begin{aligned} L_0^2 &= \left(\int_0^\pi \sqrt{a^2 \cos^2 t + c^2 \sin^2 t} dt \right)^2 \\ &\leq \pi \int_0^\pi (a^2 \cos^2 t + c^2 \sin^2 t) dt \\ &= \pi^2 \frac{a^2 + c^2}{2} = \pi^2 \frac{157}{160} < \pi^2. \end{aligned}$$

Thus $L_0 < \pi$ with strict slack.

We now make an arbitrarily small inward notch near the equator. For $\delta > 0$ small, put

$$h = c \sin \delta, \quad p_{\pm} = (a \cos \delta, \pm h).$$

Replace the elliptical arc between p_+ and p_- by the two line segments joining these points to

$$v = (a \cos \delta - \varepsilon, 0), \quad 0 < \varepsilon < a \cos \delta.$$

Because the ellipse is convex, the two new segments lie inside it. Revolving the modified simple meridian gives an embedded solid $B_{\delta, \varepsilon}$ obtained from the spheroid by a rotational inward notch.

Its meridian length satisfies

$$L_{\delta, \varepsilon} \leq L_0 + 2\sqrt{h^2 + \varepsilon^2},$$

so for sufficiently small δ and ε we still have $L_{\delta, \varepsilon} < \pi$. The removed volume lies inside the slab $|z| \leq h$ and the cylinder $\varrho \leq a$, hence

$$0 \leq V_0 - \text{Vol}(B_{\delta, \varepsilon}) \leq 2\pi a^2 h.$$

Since $h \rightarrow 0$ with $\delta \rightarrow 0$ and $V_0 > 4\pi/3$, the volume remains greater than $4\pi/3$ for sufficiently small δ .

Finally, the meridional cross-section is nonconvex: it contains p_+ and p_- , but their midpoint $(a \cos \delta, 0)$ lies outside the notched body because the new boundary radius at $z = 0$ is $a \cos \delta - \varepsilon$. Therefore $B_{\delta, \varepsilon}$ is nonconvex. Applying Theorem 2 with $R = \pi$ completes the proof. $\square$

Thus the qualitative content of Question 1 follows without computer assistance. The rest of the paper concerns the stronger quantitative problem of pushing the coefficient in (1) upward with a fully explicit certificate.

3 A convex benchmark

The classical Mylar balloon is the convex surface of revolution arising from the fixed-meridian-length volume-maximization problem studied by Paulsen and subsequently by Mladenov and Oprea [3, 4]. For a convex body of revolution, the intrinsic diameter equals the north–south meridian length R . Indeed, in meridional arclength coordinates (s, θ) the surface metric has the form $ds^2 + \varrho(s)^2 d\theta^2$, with $0 \leq s \leq R$. Any two points at arclengths s_1, s_2 can be joined through one of the poles with length at most

$$\min\{s_1 + s_2, 2R - s_1 - s_2\} \leq R,$$

whereas every path from the north pole to the south pole has length at least $\int |ds| \geq R$. Thus fixing intrinsic diameter is equivalent, in this convex axisymmetric class, to fixing the generating meridian length.

For completeness, the sharpness of the Mylar profile in this class admits a short global argument. Parametrize the meridian by arclength and write it as $(\varrho(s), z(s))$. For a general convex meridian these coordinate functions are Lipschitz; throughout this paragraph derivatives are understood almost everywhere. Convexity makes z monotone and ϱ increase from 0 to its maximum a and then decrease to 0. Hence

$$\frac{V}{\pi} = \int_0^R \varrho^2 \sqrt{1 - \varrho'^2} ds.$$

For every $0 \leq \varrho \leq a$, Cauchy–Schwarz gives

$$\varrho^2 \sqrt{1 - \varrho'^2} + |\varrho'| \sqrt{a^4 - \varrho^4} \leq a^2.$$

After integration and the monotonicity of the two halves of the meridian,

$$\frac{V}{\pi} \leq a^2 R - 2 \int_0^a \sqrt{a^4 - u^4} du = a^2 R - 2K a^3, \quad K := \int_0^1 \sqrt{1 - u^4} du.$$

The right-hand side is maximized at $a = R/(3K)$. Since $K = 2I/3$ for the constant I below, this is $a = R/(2I)$. Equality in Cauchy–Schwarz requires, on each monotone half of the meridian,

$$|\varrho'| = \sqrt{1 - (\varrho/a)^4}, \quad \sqrt{1 - \varrho'^2} = \frac{\varrho^2}{a^2}.$$

Integrating the first relation from a pole to the equator and back gives

$$R = 2a \int_0^1 \frac{du}{\sqrt{1 - u^4}} = 2aI,$$

so the equality profile has exactly the maximizing value $a = R/(2I)$. This is the classical Mylar profile, and hence the upper bound is attained.

Let

$$I = \int_0^1 \frac{du}{\sqrt{1 - u^4}} = \frac{\sqrt{\pi} \Gamma(1/4)}{4\Gamma(3/4)}.$$

The Mylar volume is

$$V_{\text{Mylar}}(R) = \frac{\pi}{12I^2} R^3 = \frac{4}{3} \left(\frac{\Gamma(3/4)}{\Gamma(1/4)} \right)^2 R^3 = 0.152315527 \dots R^3. \quad (3)$$

Its generating meridian has length R , so Theorem 2 gives a direct 1-Lipschitz wrap by the single disk D_R . This uses the Mylar balloon only as a target body; it does not identify the source with the doubled circular sheets used in the classical physical construction.

For comparison, the sphere of radius R/π also has north–south meridian length R , and hence is wrapped by D_R by the same lemma. Its volume is

$$V_{\text{sph}}(R) = \frac{4}{3\pi^2} R^3 = 0.1350949115 \dots R^3.$$

Thus the explicit benchmarks are

body	volume divided by R^3
sphere	0.1350949115 ...
sharp convex body-of-revolution Mylar benchmark	0.1523155270 ...
certified nonconvex body of this paper	0.1798434942 ...

The certified construction therefore exceeds the Mylar benchmark by about 18.07%. This is only a benchmark within the convex axisymmetric class; no optimality among all convex wrappable bodies is asserted.

4 The source complex and the one-pleat seam

It suffices to construct the case $R = 1$; scaling source and target by R then multiplies volume by R^3 .

Let H be the regular decagon inscribed in the unit circle, with outer vertices

$$o_k = (\cos(2\pi k/10), \sin(2\pi k/10)), \quad k = 0, \dots, 9.$$

Set

$$\rho = \frac{13}{20}, \quad i_k = \rho o_k,$$

and let $z = (0, 0)$ be the center. Triangulate H by the 30 triangles

$$\begin{aligned} (z, i_k, i_{k+1}), \\ (i_k, o_k, o_{k+1}), \\ (i_k, o_{k+1}, i_{k+1}), \end{aligned}$$

with indices modulo 10. All squared source lengths lie in $\mathbb{Q}(\sqrt{5})$.

The ten outer vertices are identified in cyclic order as

$$A_2, A_1, J, C, J, B_1, B_2, B_1, J, A_1. \quad (4)$$

Thus the boundary traces the graph

$$A_2 - A_1 - J - C - J - B_1 - B_2 - B_1 - J - A_1 - A_2.$$

Every seam edge is traversed twice in opposite directions, and the branch $J - C - J$ is the single pleat. The ten inner vertices and the center remain distinct. Hence the quotient has 17 vertices.

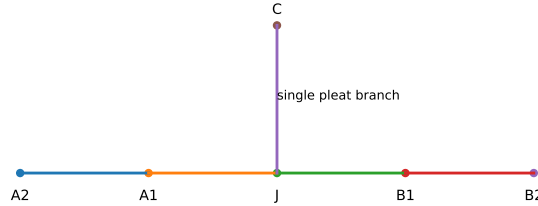


Figure 2: Combinatorial seam graph. The branch JC is the single pleat. The actual target coordinates are not coplanar and are listed in section 5.

5 Explicit target coordinates

Let

$$\lambda = \frac{99999}{100000}.$$

Each target vertex is λ times the finite-decimal vector in table 1. Every displayed decimal is interpreted as an exact rational number. The affine image of each source triangle is determined by the images of its three vertices.

Certified one-pleat construction from multiple viewpoints

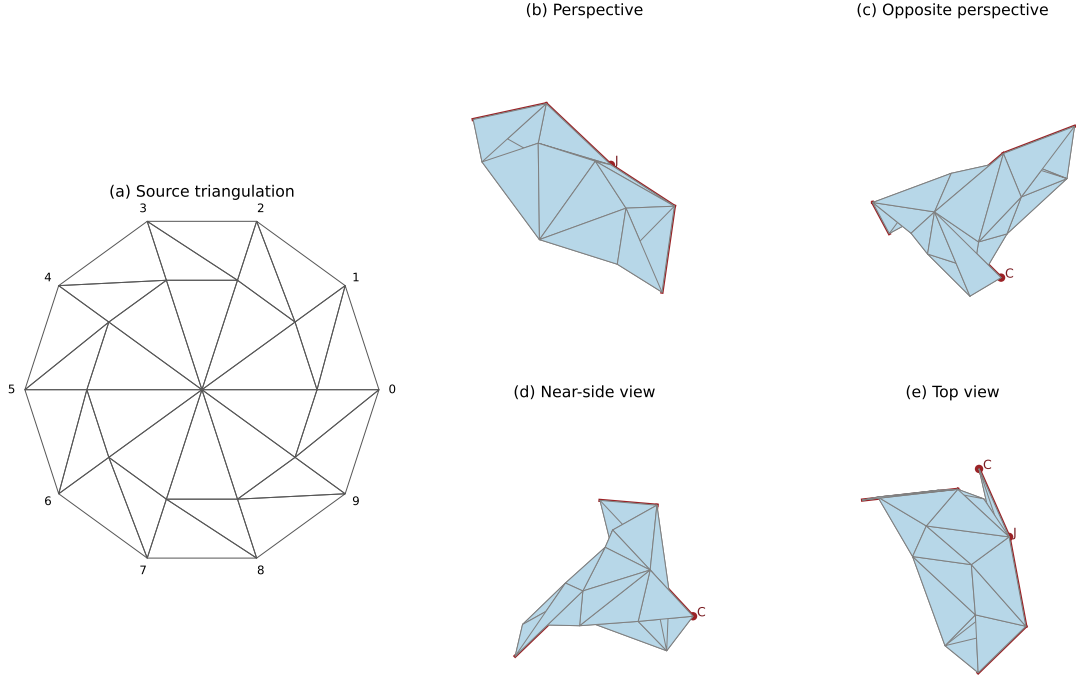


Figure 1: The source triangulation and four opaque views of the certified target surface. Red edges indicate the seam graph; the points C and J are marked in the target views. In particular, the opposite, near-side, and top views make clear that C is a vertex of the polyhedral surface, not an interior point. The drawings are for visualization only; the proof uses the exact coordinates in section 5.

Table 1: Base target coordinates before multiplication by $\lambda = 99999/100000$.

vertex	x	y	z
J	0.114996453431	0.652042463578	0.359274110539
A_1	0.713527595885	0.556758865942	0.238247176413
A_2	0.904510611077	0.150919970833	-0.186942534628
B_1	-0.269259146678	0.427202794161	0.787947375499
B_2	-0.411232941130	-0.173302559313	0.822628398965
C	-0.407447365206	0.632905197995	0.029649729336
z	0	0	0
I_0	0.650000000000	0	0
I_1	0.525861046344	0.382060413990	0
I_2	0.201227167335	0.617682813279	0.021807551587
I_3	-0.138361314859	0.605243587298	-0.192448295874
I_4	-0.192247060016	0.586039489637	0.205179883277
I_5	-0.332154604650	0.312020954927	0.463482731389
I_6	-0.424437564120	-0.078307526511	0.486025395896
I_7	-0.126467920004	0.156558885543	0.618057586773
I_8	0.208351084601	0.348929212409	0.507285156764
I_9	0.525861046344	0.263795278601	0.276373318043

Let $g : H \rightarrow \mathbb{R}^3$ denote the resulting continuous piecewise-affine map. Continuity holds because adjacent triangles use the same target coordinates on their common vertices, including the identifications in (4).

6 Exact certification

We now establish the geometric properties of g . All sign and intersection assertions in this section are checked by exact arithmetic in the supplied programs; the verification methods are described explicitly so that the certificate is finite and reproducible.

6.1 Strict nonexpansion on every triangle

For a source triangle choose two edge vectors u, v from a common vertex and set

$$S = \begin{pmatrix} \langle u, u \rangle & \langle u, v \rangle \\ \langle u, v \rangle & \langle v, v \rangle \end{pmatrix}.$$

Let T be the corresponding Gram matrix of the target edge vectors. A tangent vector with coordinate column $\xi \in \mathbb{R}^2$ has squared source length $\xi^T S \xi$ and squared target length $\xi^T T \xi$. Thus the affine triangle map is strictly nonexpansive if

$$S - T \succ 0.$$

Lemma 4. *For each of the 30 source triangles, $S - T$ is positive definite. Hence g is 1-Lipschitz on H .*

Proof. For a real symmetric 2×2 matrix, positive definiteness is equivalent to positivity of the two diagonal entries and of the determinant. The verifier computes, for each triangle,

$$(S - T)_{11} > 0, \quad (S - T)_{22} > 0, \quad \det(S - T) > 0.$$

The source terms lie in $\mathbb{Q}(\sqrt{5})$ and the target terms are rational; signs in $\mathbb{Q}(\sqrt{5})$ are reduced to rational comparisons by squaring only after the signs of the rational coefficients are known. Thus no floating-point sign decision is used.

To pass from triangles to the whole decagon, take $x, y \in H$. Convexity of H implies that the segment $[x, y]$ remains in H and crosses finitely many source triangles. On each subsegment the image curve has length at most the source subsegment length. Summing yields

$$|g(x) - g(y)| \leq \text{length}(g([x, y])) \leq |x - y|.$$

□

6.2 The quotient is a sphere

Lemma 5. *The quotient triangulation is a connected closed orientable triangulated 2-manifold with*

$$V = 17, \quad E = 45, \quad F = 30,$$

and hence is topologically S^2 .

Proof. The exact combinatorial verifier checks that every edge belongs to exactly two faces, every vertex link is one cycle, the complex is connected, and the listed face orientations agree oppositely along each shared edge. Hence it is a connected closed orientable 2-manifold. Its Euler characteristic is

$$17 - 45 + 30 = 2,$$

so the classification of closed orientable surfaces gives S^2 .

□

6.3 No unintended triangle intersections

Lemma 6. *The geometric realization of the quotient complex is embedded in $\mathbb{R}^3$.*

Proof. There are $\binom{30}{2} = 435$ unordered face pairs. The independent verifier tests intersection by solving exactly

$$A_0 + \alpha(A_1 - A_0) + \beta(A_2 - A_0) = B_0 + \gamma(B_1 - B_0) + \delta(B_2 - B_0)$$

under the barycentric inequalities

$$\alpha, \beta, 1 - \alpha - \beta, \gamma, \delta, 1 - \gamma - \delta \geq 0.$$

The three coordinate equations have rank 3 in every tested pair, leaving one affine parameter. The six inequalities become rational interval constraints on that parameter.

The exhaustive outcome is:

prescribed common simplex	number of pairs	exact intersection
none	260	empty
one vertex	130	that vertex only
one edge	45	that edge only

Thus there are no unintended self-intersections. Together with Lemma 5, the realization is an embedded PL 2-sphere. By Jordan–Brouwer separation it bounds a compact solid, denoted B_1 . $\square$

6.4 Nonconvexity

Lemma 7. *The bounded solid B_1 is nonconvex.*

Proof. Consider the face (z, I_0, I_1) and let n be its oriented normal. Exact rational arithmetic gives

$$n \cdot (J - z) = \frac{178438386772277185863216936460667247207}{200000000000000000000000000000000000000} > 0,$$

whereas

$$n \cdot (A_2 - z) = -\frac{23211889835379489551564573316040740141}{500000000000000000000000000000000000} < 0.$$

Thus vertices of the boundary lie on both sides of the plane of a boundary triangle. Every face plane of a convex polyhedron is supporting, so this is impossible for a convex solid. $\square$

6.5 Exact volume

Lemma 8. *The volume enclosed by the target sphere equals the constant C in (2).*

Proof. The face orientations are globally consistent. Therefore the oriented tetrahedral formula gives

$$\text{Vol}(B_1) = \frac{1}{6} \left| \sum_{(a,b,c) \in \mathcal{F}} \det(a,b,c) \right|.$$

All target coordinates are rational. Exact evaluation yields

$$\text{Vol}(B_1) = \frac{359686988325832312742347877865122384142690317017149}{200},$$

which is $0.17984349416291615\dots > 0.179843$.

7 From the decagon to the disk

The preceding construction uses only the regular decagon H , but the statement concerns the entire disk. The transfer is immediate from metric projection.

Lemma 9. *Let $\pi_H : D_1 \rightarrow H$ be Euclidean nearest-point projection. Then π_H is 1-Lipschitz and onto. Consequently*

$$F_1 := g \circ \pi_H : D_1 \rightarrow \mathbb{R}^3$$

is 1-Lipschitz and has image ∂B_1 .

Proof. For $x, y \in D_1$, set $p = \pi_H(x)$ and $q = \pi_H(y)$. The standard projection inequalities for a closed convex set are

$$(x - p) \cdot (q - p) \leq 0, \quad (y - q) \cdot (p - q) \leq 0.$$

Adding gives

$$|p - q|^2 \leq (p - q) \cdot (x - y) \leq |p - q| |x - y|,$$

and hence $|p - q| \leq |x - y|$. The projection fixes every point of H , so it is onto. Therefore F_1 is 1-Lipschitz and

$$F_1(D_1) = g(H) = \partial B_1.$$

□

Lemma 10 (Boundary self-gluing). *The restriction $\pi_H|_{\partial D_1} : \partial D_1 \rightarrow \partial H$ is onto. Under $F_1 = g \circ \pi_H$, every relative interior point of a seam edge has at least two distinct preimages on ∂D_1 , coming from the two oppositely oriented boundary edges in (4). Thus no boundary arc remains unglued.*

Proof. Let p lie in the relative interior of an edge e of the inscribed decagon H , and let n_e be its outward unit normal. Because p lies strictly inside the unit circle, the ray $p + tn_e$, $t \geq 0$, meets ∂D_1 at some point x . Since $x - p$ lies in the normal cone of the convex set H at p , nearest-point projection satisfies $\pi_H(x) = p$. At a vertex $p = o_k$ we simply have $p \in \partial D_1$ and $\pi_H(p) = p$. Hence $\pi_H(\partial D_1) = \partial H$.

The cyclic word (4) traverses each of the five seam segments A_2A_1 , A_1J , JC , JB_1 , and B_1B_2 exactly twice, once in each direction. On each outer decagon edge, g is affine and maps that edge onto the corresponding seam segment. Therefore a relative interior point y of any seam segment has preimages p, p' in the relative interiors of two distinct decagon edges. By the first paragraph there are $x, x' \in \partial D_1$ with $\pi_H(x) = p$ and $\pi_H(x') = p'$, so $x \neq x'$ and $F_1(x) = F_1(x') = y$. At the terminal seam vertices A_2 , C , and B_2 , the two paired boundary arcs meet at the same image vertex; at the intermediate vertices A_1 , J , and B_1 , the incident paired arcs meet according to the cyclic word (4). In every case the corresponding quotient vertex link, checked in Lemma 5, is a circle rather than an interval. Thus the relative interiors are paired and the endpoints close up without leaving boundary, so the whole disk boundary is self-glued in the image. □

Scaling both the planar domain and the target by R defines F_R and gives $\text{Vol}(B_R) = R^3 \text{Vol}(B_1)$. Together, Lemmas 4 to 10 prove Theorem 1.

8 Verification protocol and reproducibility

The accompanying repository contains two independent exact verifiers together with a `README.md` giving the exact commands, dependencies, and expected certificate output.

Verifier A

`certificate/exact_certificate.py`.

Target coordinates use Python’s `Fractions.Fraction`; source metric expressions use exact symbolic arithmetic in $\mathbb{Q}(\sqrt{5})$. Triangle intersections are tested by exact plane-section interval arithmetic.

Verifier B

`certificate/independent_verify.py`.

This implementation does not use SymPy. It contains a hand-written $\mathbb{Q}(\sqrt{5})$ arithmetic type, checks orientability by propagation on the dual graph, and tests triangle intersections by the barycentric feasibility method described in Lemma 6.

Both programs independently return the same exact rational volume and the same intersection counts

$$260 + 130 + 45 = 435.$$

A successful run prints, among other diagnostics,

```
V,E,F = 17 45 30 Euler = 2
triangle pairs = {1: 130, 2: 45, 0: 260}
volume decimal = 0.17984349416291615
all 30 triangle maps are strictly 1-Lipschitz
```

The scale factor $\lambda = 0.99999$ is intentionally conservative. It creates a strict nonexpansion margin while preserving the combinatorial and intersection structure under uniform scaling. The theorem makes no optimality claim about the displayed solid.

9 Discussion

The main conclusion is the certified lower bound

$$W(R) \geq CR^3, \quad C = 0.17984349416291615\dots$$

The construction does not identify $W(R)$ and makes no claim that one pleat is optimal. It also does not determine the supremum restricted to convex wrappable bodies. The Mylar comparison is therefore deliberately modest: it shows that the certified witness is quantitatively well beyond the sharp convex body-of-revolution benchmark, not that it beats every convex body wrappable by a disk.

The soft argument in section 2 changes the interpretation of Nandakumar’s Question 1. Under the stated 1-Lipschitz formalization, nonconvexity by itself is easy to introduce once one has a rotational wrap with strict meridian-length and volume slack. A tiny inward notch already gives an affirmative answer. The more substantive extremal question is how much volume can be enclosed, and the exact coefficient C should be viewed as a concrete lower bound for that problem rather than as a certificate needed merely to establish nonconvex existence.

The use of a piecewise-affine source map is distinct from the problem of finding a neat isometric origami folding. Work of Demaine and Ku shows that, for polygonal boundaries folded nonexpansively at finitely many points, compatible isometric mappings of the polygon interior exist under their model [5]. The present proof does not invoke that theorem: the full non-neat 1-Lipschitz map is written down directly and independently certified for self-intersection.

Natural next problems are to optimize the present crease network, to search for larger certified coefficients using finer source complexes, and to understand the best convex body wrappable by a disk. Any improvement of the first two types would raise the explicit lower bound on $W(1)$; the convex problem would clarify how much of the gain is genuinely tied to nonconvexity.

Acknowledgements

The coordinate configuration was found by computer-assisted geometric search. The final proof was reorganized into a finite exact certificate and checked by two independent implementations. External verification of both the mathematical interpretation and the code is encouraged.

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